



Yasashwini Gutta

📍 Home : Aachen, Germany

✉ Email: guttayasashwini@gmail.com 📞 Phone: (+49) 1714724951

🌐 LinkedIn: [linkedin.com/in/yasashwini-gutta](https://www.linkedin.com/in/yasashwini-gutta)

👤 GitHub: <https://github.com/yasashwinigutta>

📅 Date of birth: 31 Jul 2001

EDUCATION AND TRAINING

[Sep 2025 – Current] **Master of Science: Robotic Systems Engineering**

RWTH Aachen University, Aachen, Germany

Final grade: 2.2 (upto sem 1)

High Potential Student (HiPS Grant) Scholarship Recipient

[Jul 2019 – Apr 2023] **Bachelor of Technology: Mechanical Engineering**

National Institute of Technology Karnataka, Surathkal, India

Final grade: CGPA: 8.77/10

[May 2016 – Mar 2019] **Central Board of Secondary Education (CBSE)**

Sri Chaitanya Techno School, Bangalore, India

Final grade: 12th Class Percentage: 89.6%; 10th Class CGPA: 10/10

WORK EXPERIENCE

[Aug 2023 – Jun 2025] **Design Engineer – Chassis Motorcycles (R&D)**

TVS Motor Company Limited, Hosur, India

- Developed and standardized a **Foldable Rider Footrest** across commuter motorcycles, achieving significant annual cost reduction.
- Conceptualized an **innovative Centre Stand design**. Internal application for **patent** approved, filing in progress through the company.
- Engineered a new **Swingarm** for a premium motorcycle, gaining hands-on experience in motorcycle dynamics and end-to-end vehicle design.

[May 2022 – Jul 2022] **Intern – Advanced Engineering Group (R&D)**

TVS Motor Company Limited, Hosur, India

- Modelled a **virtual sensor** in MATLAB/SIMULINK to **predict brake pad wear** using existing vehicle sensor data from the CAN bus data logs.
- Designed a system to differentiate friction braking vs. engine braking, enabling real-time wear monitoring **without additional hardware**.

[Oct 2020 – Feb 2022] **Research Assistant under Dr N. Gnanasekaran, Associate Professor**

Mechanical Department, NITK Surathkal, India

- Investigated heat transfer in microchannel heat sinks and developed an **Artificial Neural Network (ANN)** model to **predict thermal performance**, reducing reliance on complex thermal solvers.
- Presented the work at ATE - HEFAT 2021 - 15th International Conference. ([Link to Proceedings](#))

PROJECTS

[Aug 2023 – Sep 2023] **Design and Development of a Functional Drone with Payload**

Orientation Capstone Project at TVS Motor Company

- Developed a **drone to fly autonomously** and carry 1kg payload stably at a height of 30m with **image recognition** and **obstacle avoidance** features.
- Integrated a flight controller, microprocessor and receiver to remotely control the propeller speeds with power distribution based on requirements.

[Jul 2022 – Apr 2023]

Co-operative Control of Heterogeneous Multi-UAV/UGV System

Bachelor's Thesis (Major Project)

- Developed a cooperative **UAV-UGV control system** for search and rescue operations, integrating **ROS2-based SLAM and LiDAR** for real-time target and obstacle detection.
- Implemented advanced **path-planning algorithms** in Simulink, enabling UGV navigation to targets using UAV-generated inputs.

EXTRACURRICULARS

[May 2026]

Competitive Programming & Bot Strategy Hackathon

Bonding x Itestra, Aachen

- Secured **1st place** in a **bot-programming hackathon** by developing autonomous **game strategies** for a **dynamic multiplayer environment**.
- Implemented and optimized real-time path planning, opponent prediction, and resource-management algorithms under continuously evolving game rules.

[Nov 2025 – Current]

Party AG - Studentwohnheim - Walter Eilender Haus (WEH)

Studierendenwerk, Aachen

- **Co-organized** the annual “WEHnachten” Christmas party, delivering a **successful and profitable dormitory event** through effective team coordination.
- Elected as **lead member** for the “Neueinziehers Project” event for newcomers in 2026.

[Mar 2020 – Feb 2023]

Aerodynamicist and Composites Manufacturing Lead

FSAE Race Car, Team NITKRacing, NITK Surathkal

- Designed and tested an **aerodynamic package** for a **Formula Student EV** to improve performance and cornering stability through wind tunnel analysis.
- **Led the composites manufacturing** team in developing the car's fiberglass bodywork and researched **bio-composites**, earning an honourable mention at Formula Bharat 2021.

[Jun 2022 – Oct 2022]

Throwball Team Member

Mechanical Engg. Department - Women's Throwball Team, NITK Surathkal

- Overall **Championship Winner** of Inter-Department Sports Competition 2022 (Coliseum).

LANGUAGES

German: B1.2, English: C1, Hindi: C1, Telugu: C2, Kannada: C2

SKILLS

Software and Tools

PTC CREO | ANSYS | FUSION 360 | CATIA V5 | SOLIDWORKS | AutoCAD | Simscape

Technical Skills

Python | MATLAB & SIMULINK | C/C++ | MS Office

CERTIFICATIONS

[Sep 2024]

Advanced Professional Development Program in Vehicle dynamics

[Jan 2021]

Python for Data Science, Data Analysis with Python & Data Visualization with Python

[May 2020]

Intro to Digital Manufacturing with Autodesk Fusion 360

ACHIEVEMENTS

[Dec 2018]

Karnataka Regional Mathematics Olympiad (KRMO) - Merit Scholar

[Jan 2017]

National Talent Search Examination (NTSE) Scholarship recipient, Karnataka State Rank - 20

HOBBIES AND INTERESTS

Reading Novels, Travelling, Trekking, Strength Training, Badminton